Cycle	State	PC	IR	acc	addr	mem_data_out	Notes
1	S_FETCH1	0	-	0x00	0x00	stale	begin capturing ram[0x00]
2	S_FETCH2	0	-	0x00	0x00	0x10	**ir←0x10 (LOAD 0x10); pc←1**
3	S_EXECUTE	1	0x10	0x00	0x10	stale=0x10	decode LOAD; begin capturing ram[0x10]
4	S_WRITEBACK	1	0x10	0x00	0x10	0x0F	**acc←0x0F (=A); → S_FETCH1**
5	S_FETCH1	1	0x10	0x0F	0x01	stale=0x0F	begin capturing ram[0x01]
6	S_FETCH2	1	0x10	0x0F	0x01	0x31	**ir←0x31 (ADD 0x11); pc←2**
7	S_EXECUTE	2	0x31	0x0F	0x11	stale=0x31	decode ADD; begin capturing ram[0x11]
8	S_WRITEBACK	2	0x31	0x0F	0x11	0x09	**acc←0x0F+0x09=0x18; → S_FETCH1**
9	S_FETCH1	2	0x31	0x18	0x02	stale=0x09	begin capturing ram[0x02]
10	S_FETCH2	2	0x31	0x18	0x02	0x32	**ir←0x32 (ADD 0x12); pc←3**
11	S_EXECUTE	3	0x32	0x18	0x12	stale=0x32	decode ADD; begin capturing ram[0x12]
12	S_WRITEBACK	3	0x32	0x18	0x12	0xFC	**acc←0x18+0xFC=0x14; → S_FETCH1**
13	S_FETCH1	3	0x32	0x14	0x03	stale=0xFC	begin capturing ram[0x03]
14	S_FETCH2	3	0x32	0x14	0x03	0x53	**ir←0x53 (STORE 0x13); pc←4**
15	S_EXECUTE	4	0x53	0x14	0x13	stale=0x53	decode STORE; begin capturing ram[0x13]
16	S_WRITEBACK	4	0x53	0x14	0x13	0x00	**mem_we=1: ram[0x13]←acc=0x14; → S_FETCH1**
17	S_FETCH1	4	0x53	0x14	0x04	stale=0x00	begin capturing ram[0x04]
18	S_FETCH2	4	0x53	0x14	0x04	0x60	**ir←0x60 (HALT); pc←5**
19	S_EXECUTE	5	0x60	0x14	-	stale=0x60	decode HALT; → S_HALT
20	S_HALT	5	0x60	0x14	-	-	**done='1' (loops here) Yes, there are 20 cycles, not 19, as may be seen in waveform**